



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to NPS PETROLEUMS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

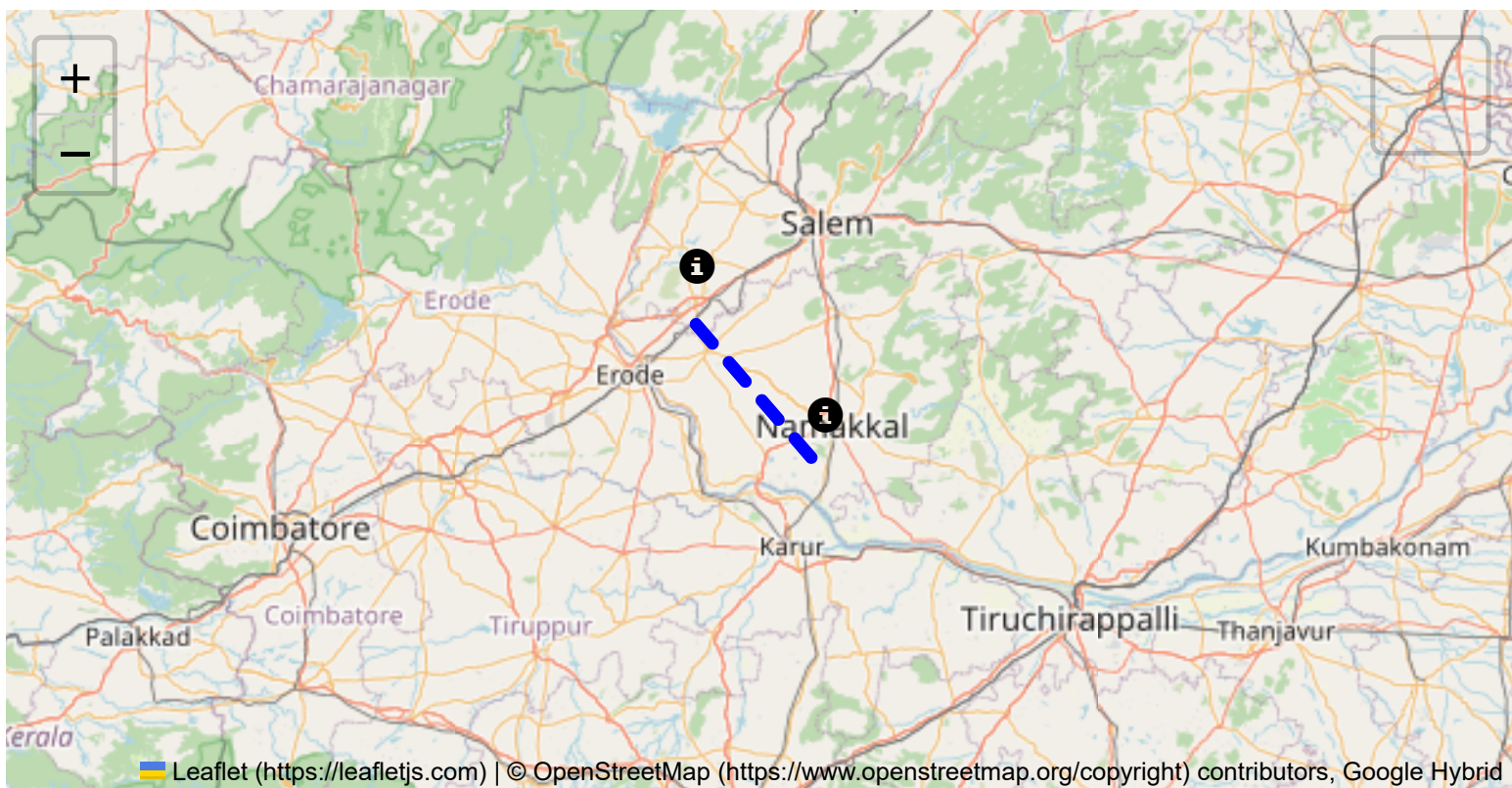
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

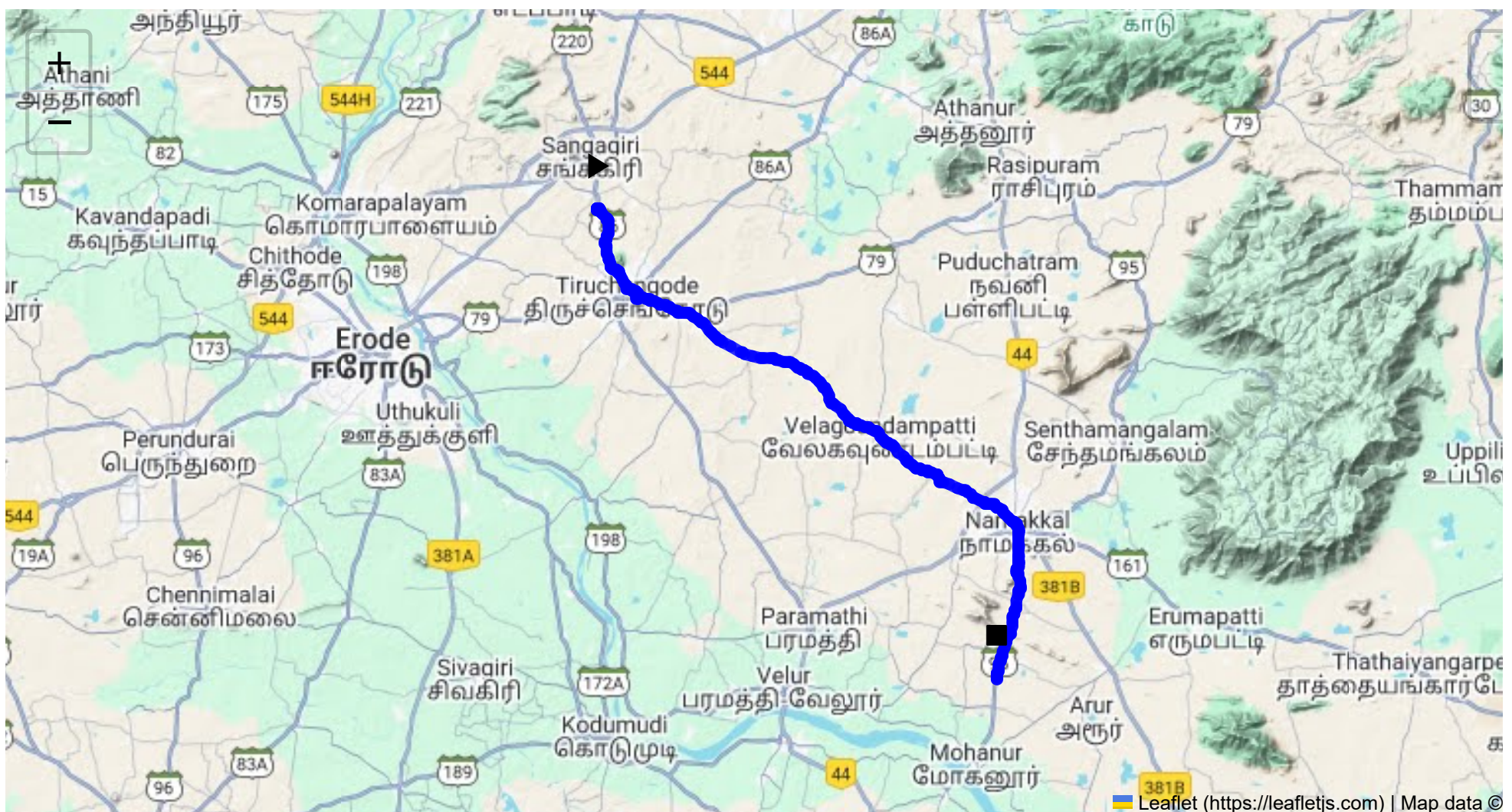
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.16 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.117262, 78.151831)



Welcome to the Journey Risk Management Study

Analyzing the route from Sangagiri, Tamil Nadu to NeikkaranPatti, Tamil Nadu involves several safety considerations due to its approximate 56.16 kilometers span, often requiring around 1.28 hours for trucks transporting hazardous materials. Here is an extensive review of potential hazards and other important factors:

1. Overview of the Route Map:

- The route primarily traverses rural and semi-urban areas, with highways and smaller roads forming the main corridors. The primary highway segments likely include national highways and state highways, interspersed with local roads leading into NeikkaranPatti.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu typically experiences hot and humid conditions but is also subject to monsoons from June to September and potential cyclonic activity. During the monsoon, the region can experience heavy rains that may lead to flooding and landslides, especially in less developed areas.

3. Analysis of Traffic Patterns:

- Traffic is generally lighter in rural areas but can become congested near towns and junctions. Peak hours usually align with local business hours (8 AM to 10 AM, and 5 PM to 7 PM). Roads nearer to commercial areas or marketplaces are prone to congestion during these periods.

4. Assessment of Road Quality and Infrastructure:

- Major highways are usually of reasonable quality, with potential potholes and uneven surfaces posing risks on smaller roads. Signage and lighting can be inadequate in rural sections, affecting visibility at night or during adverse weather conditions.

5. Suggestions for Alternative Routes for Emergencies:

- Possible detours could include using state highways or alternative regional roads, though these may cover additional distances. It's important to review local maps and have GPS-equipped systems for real-time navigation.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Regulations include specific timing for transport to avoid peak traffic, mandatory signage on vehicles, and adherence to speed limits. Transporting hazardous materials during night hours might be restricted.

7. Overview of Historical Incidents:

- While specific local data might not be widely available, heavy vehicle incidents often involve overturns due to sharp turns or collisions in low-visibility conditions. Historical data generally points to human error and vehicle conditions as primary causes.

8. Environmental Considerations and Sensitive Areas:

- Routes passing near agricultural areas or close to water bodies must be carefully navigated to avoid contamination. Wildlife crossings can also be an issue, especially at dawn or dusk.

9. Analysis of Communication Coverage:

- Rural areas may have sporadic cellular network coverage, with potential dead zones especially likely in more remote farming or forested sections.

10. Estimated Emergency Response Times:

- Response times in urban areas can typically be below 30 minutes, while rural sections might exceed 45 minutes to an hour due to distance from medical or fire services.

11. Overall Summary of Risk Assessment:

- The risk level for transporting hazardous materials on this route is moderate. Key risks involve weather-related complications, road infrastructure, and communication gaps. Ensuring vehicle readiness, driver training, adherence to regulations, and having emergency protocols in place can mitigate these risks effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	Medium	11.38342, 77.89975	30 KM/Hr
5	Turn	Medium	11.38295, 77.90001	30 KM/Hr
6	Turn	High	11.38186, 77.90258	15 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
8	Turn	Medium	11.22407, 78.16640	30 KM/Hr
9	Blind Spot	Blind Spot	11.21912, 78.16824	10 KM/Hr
10	Turn	Medium	11.21889, 78.16777	30 KM/Hr
11	Turn	Medium	11.20117, 78.16779	30 KM/Hr
12	Turn	Medium	11.19145, 78.16640	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
20	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
21	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
22	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
23	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
24	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
25	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
26	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
27	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
28	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
29	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
31	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
32	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
33	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
34	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
45	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
46	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
47	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
48	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
49	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
50	hospital	Chellam Hospitals	11.2153614, 78.1684456	30 km/h	Medium
51	hospital	Deepthi Hospital	11.2152757, 78.1680221	30 km/h	Medium
52	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium
53	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
55	hospital	Sabari Childrens Hospital	11.2151465, 78.1680375	30 km/h	Medium
56	hospital	Sri Ragavendra Hospital	11.2151734, 78.1703431	30 km/h	Medium
58	hospital	R.K. Hospital, Namakkal	11.2143252, 78.1678846	30 km/h	Medium
59	hospital	Cm Hospital	11.2136963, 78.1680022	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
60	hospital	C.M. Speciality Hospital and Jaya Lizarov Centre	11.2122571, 78.1697107	30 km/h	Medium
61	hospital	Maharaja Speciality Hospital	11.210755, 78.166997	30 km/h	Medium
62	hospital	Kiruba Hospital	11.2107891, 78.1677358	30 km/h	Medium
63	clinic	Sri Swastha Psychiatric Clinic	11.2085069, 78.1677043	30 km/h	Medium
65	hospital	Pragalath Hospital	11.2022482, 78.1679145	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
17	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
30	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium
43	school	Government Girls Higher Secondary School	11.2179186, 78.1672383	30 km/h	Medium
54	school	Namakkal Government Boys Higher Secondary School	11.2150472, 78.1669273	30 km/h	Medium
57	school	Kandasamy Kandar Matriculation and Higher Secondary School	11.2137988, 78.1698234	30 km/h	Medium
64	school	Kondichettipatti School	11.200963, 78.165767	30 km/h	Medium
66	school	Jai Vikas Higher Secondary School	11.1841646, 78.1664109	30 km/h	Medium
67	school	Trinity Academy School	11.1790291, 78.166089	30 km/h	Medium
68	college	Arignar Anna Government Arts College for Men	11.1728714, 78.1657512	30 km/h	Medium
69	university	Tamil Nadu Veterinary and Animal Sciences University	11.1589061, 78.1597079	30 km/h	Medium
70	school	NSAU Govt School	11.1420516, 78.1595648	30 km/h	Medium
71	school	Little Angle School	11.1430354, 78.1574011	30 km/h	Medium
72	college	SRG Engineering College	11.133654, 78.1532385	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38295, 77.90001



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22407, 78.16640



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.21912, 78.16824



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.21889, 78.16777



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.20117, 78.16779



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.19145, 78.16640

